

# Sergio Fuentes Cámara

*Computer Vision Engineer - ML Edge Computing Specialist*

Madrid, España +34 666 113 180 [s.fuentescamara@gmail.com](mailto:s.fuentescamara@gmail.com)  
 Portfolio: [myportfolio-mwfs.onrender.com](https://myportfolio-mwfs.onrender.com) Github LinkedIn

## Professional Profile

Computer Vision Engineer with +5 years specialising in real-time AI systems for edge devices. Experience implementing detection, OCR and tracking models in production processing +20 FPS in Full HD. Stack: Python, TensorFlow, PyTorch, Docker, FastAPI. Currently at Etra Group optimizing models for NPUs

## Professional Experience

## Computer Vision Engineer — Grupo Etra

2023 - Present

- Architected and implemented end-to-end Computer Vision pipeline in C++/Python processing Full HD video in real-time (<50ms latency) on NPU edge devices, reducing hardware costs by 70% vs. GPU solutions
- Designed and optimised custom object detection and OCR models achieving 97% accuracy in production, processing +15K operations/day through fine-tuning and transfer learning
- Led DevOps infrastructure for fleet of +20 edge devices implementing CI/CD with Docker, Ansible, and centralized monitoring, reducing deployment time by Nx where N is the number of devices
- Developed full-stack web platform (FastAPI, WebSocket, WebRTC) managing +10 simultaneous streams in real time, reducing remote configuration time by 5x

## Software Engineer — Capgemini España SL

2022 - 2023

- Developed Computer Vision system for industrial quality control automating metric measurement in production line, improving precision and reducing inspection times
- Implemented CI/CD pipelines with DevOps methodology (Git, Docker, Jenkins) for aeronautical project, ensuring continuous delivery and reducing production errors
- Designed scalable and modular software architectures following SOLID principles and design patterns, facilitating code maintainability and extensibility
- Built data preprocessing pipelines with NumPy and Pandas to feed ML models, optimizing performance and prediction quality

# Electronics Engineer and Firmware Developer — Digitanimal

2020 - 2022

- Designed and programmed low-power microcontroller firmware in C for agricultural IoT products, optimizing battery autonomy and GSM/LoRa/Sigfox communication protocols
- Developed sensor processing algorithms (GPS, accelerometer, temperature) and efficient data transmission for real-time livestock monitoring systems
- Designed complete electronic circuits including schematics, PCBs, and prototype validation for GPS collars and self-weighing scales in harsh industrial environments
- Led full prototyping cycle from mechanical 3D design (SolidWorks) to manufacturing with 3D printing (FDM/resin), rapidly iterating to final product
- Contributed to R&D of new IoT products evaluating technical feasibility, developing POCs, and accelerating time-to-market for hardware-software innovations

## Technical Skills

## Computer Vision & AI:

- TensorFlow & PyTorch
- Object detection & Tracking
- OpenCV
- OCR
- Model Optimization

## Edge Computing & Hardware:

- NPU acceleration
- Low-latency systems

## Backend & APIs:

- Embedded C/C++
- Microcontrollers & IoT
- Python
- FastAPI & REST APIs
- WebSockets & WebRTC

## DevOps & Cloud:

- MongoDB & SQL
- Asyncio & multiprocessing
- Docker
- Ansible
- CI/CD (Git, Jenkins)

## Frontend & Design:

- AWS & GCP
- Linux & Bash
- JavaScript & HTML/CSS
- React
- Flutter
- SolidWorks (3D design)

## Languages

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**Spanish** : Native

**English** : Technical (documentation, papers, code) — Conversational in development

## Education

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**Bachelor's Degree in Industrial Electronics and Automation Engineering**

*University of Castilla-La Mancha (UCLM), Toledo*

Specialization in Advanced Electronic Technologies

## Certificates and Courses

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**TensorFlow Developer**, *DeepLearning.AI* ([link](#))

*August 2023*

**DevOps Fundamentals Course**, *GitHub/90DaysOfDevOps* ([link](#))

*October 2022*

**AWS Computer Vision with GluonCV**, *AWS Training and Certification* ([link](#))

*July 2022*

**Image Segmentation with Python and Unsupervised Learning**, *Coursera Project Network* ([link](#))

*July 2022*

**Object Detection in Computer Vision**, *Autonomous University of Barcelona* ([link](#))

*January 2020*